

EVALUATE THE CORROSION PERCENTAGE IN CONCRETE TMT BARS USING THE WEIGHT LOSS METHOD

Exp. No. :

Date :

AIM/ OBJECTIVE:

To determine the corrosion percentage of iron rod in a given acidic solution.

PRINCIPLE:

Corrosion can be defined as the gradual destruction of a metal by unwanted chemical or electrochemical attack by its environment. An iron rod is exposed to the acid medium for a specified time and the amount of corrosion taken place in the iron rod can be estimated by titration method. The iron solution is titrated against standard potassium permanganate solution.

During redox reaction ferrous ion gets oxidized [$\text{Fe}^{2+} \rightarrow \text{Fe}^{3+}$] to ferric ion. From the amount of iron content in the corrosion media, the corrosion percentage is calculated by using the weight loss formula:

$$\text{Corrosion percentage} = \frac{\text{Amount of iron loss}}{\text{Initial weight of Iron strip}} \times 100$$

PROCEDURE:

PREPARATION OF IRON SOLUTION

50 ml of 0.1 N dilute H_2SO_4 solution is taken in a 250 ml beaker. An iron rod is immersed in the solution about 30 min as induction period. The steel rod is removed from the beaker. This solution is used as iron solution for the determination of iron corrosion rate.

DETERMINATION OF CORROSION RATE

20 ml of the given iron solution is pipetted out into a clean conical flask. The solution is titrated against standard potassium permanganate solution taken in the burette. The end point is appearance of pale pink color. The titration is repeated for concordant value.

OBSERVATIONS/ INFERENCE:**Determination of corrosion rate by weight loss method**

Initial weight of the Iron strip = ____g

Standardization of Iron solution (Std. KMnO_4 Vs Iron solution)

S.No	Volume of Iron Solution V_1 (ml)	Burette Reading (ml)		Volume of KMnO_4 consumed (ml)	Concordant value V_2 (ml)
		Initial	Final		
1					
2					

Calculation

Volume of iron solution (V_1) = 20 ml

Normality of iron solution (N_1) = N

Volume of KMnO_4 solution (V_2) = ml

Normality of KMnO_4 solution (N_2) = 0.1 N

Applying volumetric principle

$$V_1 N_1 = V_2 N_2$$

$$N_1 = V_2 N_2 / V_1$$

=

Normality of iron solution (N_1) = N

The amount of corrosion taken place

in the iron rod

= Normality $\times$ Equivalent weight of Iron

= $\times 55.85$

= g

The amount of corrosion taken place in the iron rod is = g/L.

The percentage of iron rod in the given solution = $\frac{\text{Amount of iron loss}}{\text{Initial weight of Iron strip}} \times 100$

=

= %

RESULTS & DISCUSSION:

The amount of corrosion taken place in the iron rod is = g

The percentage corrosion of iron rod in the given solution = %

MAPPING OF PO & PSO: (For all the COs covered by this experiment)

CO No	PO1	PO2
3	2	1

ASSESSMENT:

Particulars	Max. Marks	Marks awarded
Preparation	10	
Conduct of Experiment	30	
Results & Discussion	30	
Viva - Voce	20	
Report	10	
Total	100	
Evaluator's Signature		